

Effectiveness of Green Infrastructure for Urban Stormwater Management: A Case Study of Typical Land Uses in Central Phnom Penh

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Abstract

Phnom Penh, a rapidly developing tropical city with a significant proportion of impervious urban surfaces, has been vulnerable to urban flooding during the rainy season for nearly three decades. The approach of green infrastructure (GI) provides a potential solution to impervious surface in urban areas by providing source control through both pre-development and retrofitting. This study aims to investigate the effectiveness of GI features for mitigating urban stormwater runoff across different land-use types, namely residential (RES), commercial (COM), and mixed residential and commercial (MIX), in central Phnom Penh. Quantum Geographic Information System (QGIS), supported by Google Satellite overlay imagery, was used to classify land uses and land covers and to implement GI features. The Rational Method was employed to estimate peak runoff rates, and the overall results were computed and analyzed. As a result, Sites COM and MIX demonstrated high performance in reducing stormwater runoff, whereas Site RES exhibited moderate performance. Based on their weighted calculation, trees and permeable pavements exhibited the best results; green roofs had high performance; and bioswales demonstrated low performance. Overall, this study reveals that GI features significantly mitigate urban flooding problems by reducing peak runoff rates in typical land-use in tropical cities, such as Phnom Penh.

Keywords: green infrastructure, stormwater runoff, tropical climate, urban land use/land cover (LULC), Phnom Penh

Introduction

The expansion of urbanization and associated land-use changes significantly increases the proportion of impervious surfaces. The transition from natural to the built

environment, namely buildings, sidewalks, streets, and parking lots, exerts pressure on the hydrological cycle by reducing rainwater infiltration. They are increasing urban stormwater runoff volumes and flow rates, contributing to negative impacts such as

increased urban temperatures, soil erosion, and flooding, all of which adversely affect wildlife and human health (McGrane, 2016). Phnom Penh, a tropical city, experiences annual stormwater runoff issues stemming from population growth and urban development (Yen et al., 2017). Monsoons influence the city's climate by expanding the floodplain during storm events. Additionally, inefficient drainage system management, coupled with poor land-use management strategies, significantly contributes to serious environmental problems, including flooding in Phnom Penh's lowland areas (Retka, 2018).

The concept of Green infrastructure (GI) is significant for enhancing spatial planning by providing diverse ecosystem benefits in urban areas. It is a sustainable approach that uses nature-based solutions by offering numerous environmental benefits. GI can mitigate urban heat effects, improve air and water quality, adapt to energy and climate change, enhance human well-being, and provide a cost-effective solution (Herath & Bai, 2024). It particularly manages stormwater runoff through onsite control, mimicking the natural hydrological behavior of pre-developed urban environments (Keeley et al., 2013). Globally, GI features take many forms, including retention ponds, constructed wetlands, detention ponds, trees, rain gardens, bioswales/vegetated swales, green roofs, permeable pavements, and rainwater harvesting (United States Environmental Protection Agency, 2025). These features are employed to reduce the frequency and intensity of flooding events by slightly shortening peak flow duration, decreasing runoff volume, and removing

water pollutants through processes such as canopy interception, retention, infiltration, and evapotranspiration (McFarland et al., 2019). A study by Martínez et al. (2018) on potential GI applications found that GI optimally reduced peak runoff (28%), runoff volume (60%), and pollutants (33%).

While stormwater management is a major concern in cities with high annual rainfall intensity, limited research has been conducted in these tropical climates (less than 10%) (Parker & Zingoni de Baro, 2019). Especially in developing countries, flooding is more severe than in developed countries, as rapid urbanization, land-use change, and poor drainage infrastructure have led to urban flooding. Phnom Penh, a tropical city in Cambodia, annually faces stormwater runoff issues due to population growth and rapid urban development (Ahmed et al., 2026; Yen et al., 2017). The amount of rainfall or storm events creates a larger-scale floodplain in Phnom Penh. Normal Monsoon rains; flood levels reach 1.5 m in some parts of Phnom Penh during the rainy season (Doyle, 2012). Besides, the inefficiency of drainage system management, coupled with poor land-use management strategies, causes serious environmental problems, including floods in the lowland areas of Phnom Penh (Retka, 2018). Consequently, it is a crucial opportunity to estimate the runoff reduction resulting from the implementation of GI measures.

This study aims to investigate the effectiveness of green infrastructure (GI) features in de-escalating urban stormwater runoff across different land-use/land-cover types in Phnom Penh. Three different urban

land uses, namely Residential, commercial, and mixed land uses, are typical land uses for cities in developing countries. While Nou and Charoenkit (2020) investigated the implementation of green infrastructure features, namely trees, permeable pavements, bioswales, and green roofs in residential neighborhoods for stormwater runoff reduction, this study explored these features in the two other land use types, namely commercial and mixed residential and commercial land use. The findings of this study can contribute to the body of knowledge in the application of GI features in a tropical city as follows: 1) The effectiveness of combined and individual GI features in reducing runoff in typical landuses/land covers; 2) recommendations for other cities that have similar

characteristics, developing and tropical cities, by comparing GI features in different land use types.

Methodology

Study Framework

This study designed an overall framework, divided into three stages, as shown in Figure 1, to estimate the total reduction in runoff rate in central Phnom Penh. First, a literature review was conducted to identify suitable land-use types for study sites in Phnom Penh's central urban area. Understanding the interrelation between the specification of green infrastructure (GI) and the land use/land cover types is vital for assessing the overall effectiveness of GI. The performance

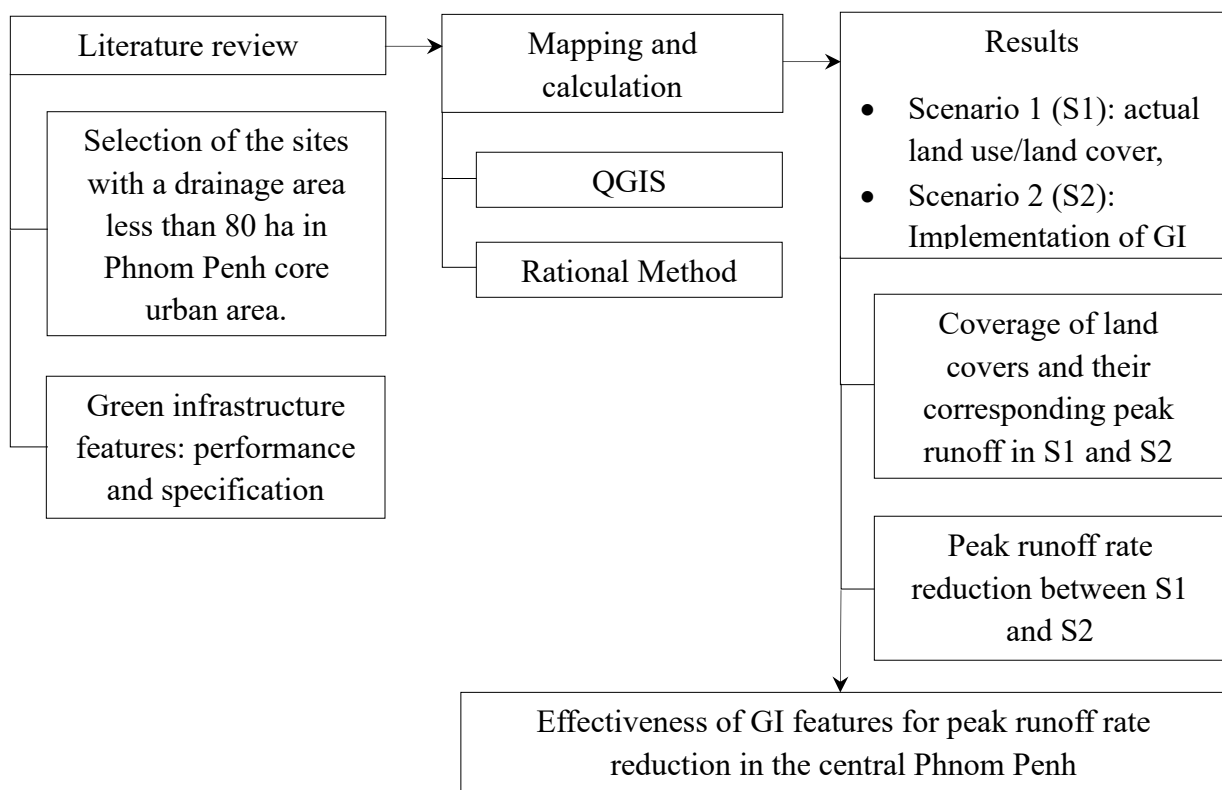


Figure 1. Overall framework of the study

and specifications of GI features were reviewed for the selected sites. Second, the Quantum Geographic Information System (QGIS) and Google satellite imagery were integrated to classify the existing land covers at each site (e.g., roofs, streets, parking lots). They were mapped and verified through Google Earth and onsite visits (in 2019). The study also used the Rational Method to compare peak runoff rates between two scenarios: the current situation without GI (Scenario 1) and a proposed situation with GI features replacing existing impervious land covers (Scenario 2). The different land uses/land covers, including perviousness, types, and sizes, are recorded in QGIS for computing. The results in S1 and S2 were then analyzed to evaluate the effectiveness of GI features to decrease peak runoff in the city.

Site Selection

Past observations in central Phnom Penh indicate that some areas remain inundated for several days or over a week. According to Wen et al. (2016), inundation was frequently observed in the central urban area across a few districts: Chamkar Mon, Daun Penh, and Prampir Meakara. While Nou and Charoenkit (2020) studied residential land use in Chamkar Mon, this study targeted commercial and mixed residential-commercial land use in the Wat Phnom and Teuk L'ak Ti Muoy communes, respectively, given their status as flood-prone communities (Figure 2). Table 1 illustrates these two selected sites—drainage basins under 80 hectares each, representing COM and MIX—and their specifications.

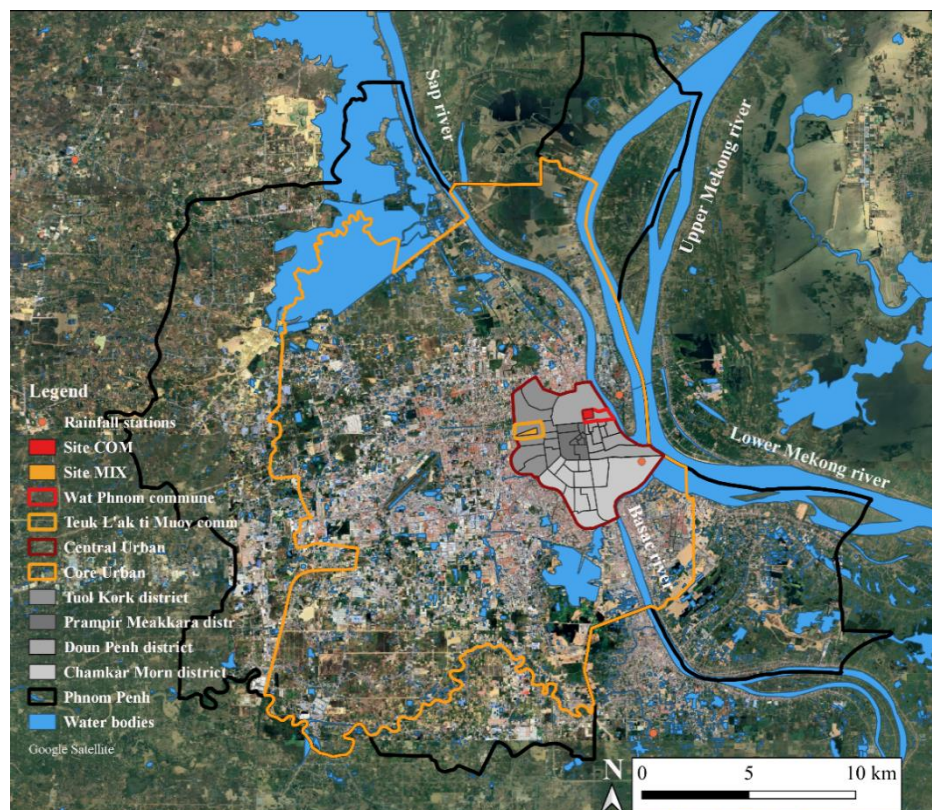


Figure 2. Map of the two sites in the Phnom Penh central urban area. (Source: Author)

Table 1. Specification of sites COM and MIX.

Site	COM	MIX
Location	11°34'20.39"N, 104°55'13.79"E, situated in Wat Phnom commune, Doun Penh district (near Tonle Sap)	11° 34' 2.18"N, 104° 53' 40.26" E, is in Teuk L'ak Muoy commune, Tuol Kouk district
Coverage	20.83 ha	19.50 ha
Density	14,378/km ²	14,858/km ²
Land-use type and its characteristic	• Medium-density urban area • Commercial and institutional areas • 70–75% impervious cover, and public green spaces	• Medium-density urban area • A combination of commercial and residential blocks • 75 - 80 % of impervious surfaces
Problems	• Inundation depth: mostly up to the knees from rainfall and lasted for two to three hours (JICA, 2016; Yim et al., 2016).	• Inundation depth: up to one meter (Hong et al., 2016) and lasts for two to three hours (JICA, 2016)

Performance and Specification of Green Infrastructure Features

Green infrastructure (GI) features appear in a variety of forms. In this study, four features are selected for implementation in Phnom Penh's core urban areas because they are flexible, adaptable, and feasible: trees, bioswales, permeable pavements, and green roofs (Nou & Charoenkit, 2020). In urban environments, trees are a common form of green infrastructure (GI). They reduce stormwater runoff and delay its impact through three processes: rainfall interception by tree canopies, infiltration through tree pits, and evapotranspiration through leaves (Asadian & Weiler, 2009; Berland et al., 2017; Huang et al., 2017). In addition, tree pits restore natural flow even in urbanized areas with infiltration soils, and substantially reduce surface runoff up to 66% (Armson et

al., 2013; Elliott et al., 2018). Tree canopies are a crucial component of the urban hydrologic cycle, as they evaporate approximately 6.5–27% of total rainfall (Kirnbauer et al., 2013). In Southeast Asia, Africa, and South America, deciduous tree species are common due to their high transpiration rates. It maximizes photosynthesis and allows them to access and store soil water during the wet season in tropical rainfall-seasonality-driven systems (Singh & Kushwaha, 2016).

Bioswales or vegetated swales are essential for capturing and treating stormwater runoff. They are frequently implemented along residential roadways and in parking lots. In a small-scale usage, they primarily comprise vegetation, substrate, and soil (Shrestha et al., 2018; Thiagarajan et al., 2018). Shrubs are often ideal for sloped areas because their root

systems can access soil moisture (Mazer et al., 2001). The National Research Council (NRC) recommends constructing bioswales on gentle slopes (up to 5%) to prevent erosion and promote rapid infiltration of rainwater. For instance, bioswales incorporating engineered soil and trees for reducing parking lot runoff by up to 88.8% in Mediterranean climates (Xiao & McPherson, 2011). In humid continental climates, these systems have been observed to capture 48–96% of storm event volumes (Shrestha et al., 2018).

Permeable pavements provide substantial environmental benefits, primarily allowing stormwater to infiltrate through their surface and underlying layers. This process effectively minimizes runoff volume and helps manage stormwater from rooftops, parking lots, sidewalks, driveways, and residential streets. Common types of permeable pavements include pervious concrete (PCs), porous asphalt (PAs), and block pavers (e.g., permeable interlocking concrete pavers (PICPs), concrete grid pavers (CGPs), and plastic reinforcement grid pavers (PRGPs). Permeable systems typically comprise a permeable surface, an aggregate base, a sub-base layer (storage), and a soil layer (exfiltration) (Fassman & Blackburn, 2010). The thickness of these components is critical, as it facilitates reducing surface runoff volume and peak flows in each layer (Collins Kelly et al., 2012). High soil permeability is commonly found in sand or native soils with less than 40% silt/clay content and less than 20% clay (Jayasooriya & Ng, 2013). For instance, a 45–55% reduction in peak discharge and a 50–60% reduction in runoff volume compared to

conventional asphalt (Dreelin et al., 2006; Terkura & Akib, 2024).

Green roofs are essential green infrastructure (GI) features, designed to mitigate stormwater runoff from rooftops. Their hydrological function encompasses the interception, retention, and evaporation of rainwater. The effectiveness of green roofs is primarily influenced by several factors, including substrate thickness, vegetation type, roof slope, and rainfall event size (Golden & Hoghooghi, 2018; Liu & Chui, 2019; Viola et al., 2017; Volder & Dvorak, 2014). There are two primary classifications: extensive green roofs and intensive green roofs. Extensive green roofs are lightweight, with substrates less than 150 mm deep, which support shallow-rooted plants such as sedum. Additionally, intensive green roofs feature thicker substrates (exceeding 150 mm) that accommodate a broader range of vegetation. However, these require more maintenance and are generally restricted to flat or gently sloped roofs (Kosareo & Ries, 2007; Mentens et al., 2006; Stovin, 2010). Many studies indicate substantial rainfall retention: extensive and intensive green roofs retain 11% and 77% of rainfall, respectively, across various global locations. Even higher retention rates (49.9% for extensive, 57.2% for intensive) have been observed in tropical climates, largely attributable to increased evapotranspiration (Ebrahimian et al., 2019; Viola et al., 2017).

Ultimately, the specification of the four GI features for implementation is summarized in Table 2. They are designed based on the relationship between appropriate land covers, sizes and their applicability within two distinct urban land uses.

Table 2. Summary of GI features specification for implementation.

GI features	Specification of the implementation in the various land covers
Trees	- Deciduous tree species - On sidewalks where the widths are wider than 1.5 m for tree pits, 8m each
Bioswales	- Along boulevards: replaced an existing concrete median barrier - In parking lots
Permeable pavements	- Sidewalks, parking lots, streets, and others - The impervious pavers, concrete, and asphalt are replaced by pervious pavers, concrete, and asphalt, respectively.
Green roofs	- Fully covered flat roofs - Intensive green roofs - Key characteristics include vegetation such as sedums, herbs, and small plants, and a substantial substrate depth of 0.15 to 1.6 meters.

Rational Method

The Rational Method is employed to calculate the peak runoff rate (Q) in urban or suburban areas resulting from a uniform rainfall intensity over a selected period (see Eq. 1). This method is simple and effective for runoff estimation with small urban watersheds with drainage areas (A) less than 80 hectares that exhibit generally uniform surface cover and topography (Rene Garcia, 2016). Hence, this method is appropriate for the selected sites due to the large amount of impervious surfaces, site size, surface cover, and topography, as well as the limited availability of data in the Phnom Penh central urban area.

$$Q = \frac{1}{360} CiA \quad (\text{Eq 1})$$

The average rainfall intensity (i) required for the Rational Method corresponds to a 2-year

storm return period, a common standard in numerous studies for estimating runoff volume. This study will use data from the Pochentong meteorological station in Phnom Penh, which indicates a 2-year storm return period of 44.8 mm/h, as it is near the selected sites.

The runoff coefficient is a value between 0 and 1 that defines the proportion of rainfall that becomes runoff. It accounts for infiltration, interception, evapotranspiration, and retention due to a surface's perviousness. Due to the lack of data available in developing countries as well as in tropical climate zones, this study uses the runoff coefficient, C and C', in scenarios 1 and 2, respectively. Besides, averaging across more than two studies per land cover can improve accuracy in this study, as found by Nou and Charoenkit (2020).

The coverage of land use/cover and Green Infrastructure (GI) features was subsequently calculated in the QGIS attribute table for both scenarios, as shown in Eq. 2.

$$\text{Coverage (\%)} = \frac{A_i}{\sum_{i=1}^n A_i} \quad (\text{Eq 2})$$

Where, A_i = Area of land cover ($i=1, 2, \dots, n$)

Peak Runoff Rate Reduction

Once all necessary input variables were obtained, peak runoff rates were computed. The percentage reduction in peak runoff rate for the site COM and MIX was subsequently determined by quantitatively comparing the peak runoff rates between the two scenarios (Q_{S1} & Q_{S2}), as illustrated in Eq. 3.

$$Q_{\text{Reduction}} (\%) = \frac{Q_{S1} - Q_{S2}}{Q_{S1}} \quad (\text{Eq 3})$$

Defining the Effectiveness of Green Infrastructure Features

The effectiveness of green infrastructure (GI) features in reducing runoff was evaluated by weighing. To get the value of Weight, the comparison between the peak runoff rate reduction ($Q_{\text{Reduction}}$) produced by individual and combined GI features, to the coverage of individual and combined GI features, respectively (Eq. 4). In addition, the coverage of individual GI features to the combined GI features is derived from the comparison between coverage of individual GI feature to the total site area and the combined GI features is a sum of individual GI feature to the total site area (obtained from the contributing table in QGIS after the implementation of GI features in S2).

$$\text{Weight} = \frac{Q_{\text{Reduction}}}{\text{Coverage of GI features}} \quad (\text{Eq 4})$$

The further assessment of their overall effectiveness was subsequently categorized into four levels based on this calculated Weight: "best" (weight ≥ 1), "high" ($0.75 \leq \text{Weight} < 1$), "medium" ($0.50 \leq \text{Weight} < 0.75$), and "low" (Weight < 0.50).

Results

Coverage of Land Use/Land Cover and Peak Runoff Rate Reduction in Site COM

Land use/land cover in the site COM predominantly features public green spaces, commercial and institutional blocks, and transportation infrastructure. A mixture of grasses, tree canopies, flat and sloped roofs, and impervious surfaces such as concrete, pavers, and asphalt characterizes the surface covers. In scenario 1, pervious cover constitutes 27.40% of the total area, while impervious cover accounts for 72.60%.

Public green spaces represent the largest share, approximately one-fifth of the total area, followed by sloped roofs (19.87%), other land covers (16.99%), streets (12.31%), and flat roofs (12.02%). Sidewalks and parking lots each contribute approximately one-fifth of the total area. In addition to the runoff rate, sloped roofs exhibited the highest peak runoff rate at 0.49 m³/s, accounting for 26.16% of the total, or approximately one-quarter. Streets, flat roofs, parking lots, and sidewalks followed in descending order. Public green spaces contributed the least to runoff, at only 2.58% (0.05 m³/s). It also had a peak runoff rate of 1.87 m³/s, with

impervious covers accounting for a third to a fourth of the total area (see Table 3).

In scenario 2, permeable pavement shows the most prevalent GI feature at this site, accounting for 37.21% of the total area when implemented on sidewalks, streets, parking lots, and other land covers. Green roofs rank second in applicability, accounting for 12.02%, due to the large share of flat roofs in buildings in this area. Trees were planted

across 10.13% of the total area, with applications on sidewalks, streets, and parking lots. Bioswales are least applicable due to limited space along streets and parking lots. It reached a peak runoff rate of 0.94 m³/s when 59.71% of GI features were implemented, and 49.63 % of peak runoff was reduced (0.92 m³/s) for a 2-year storm return period, with impervious surfaces decreasing.

Table 3 Distribution of land covers in scenario 1, implementation of GI features across different urban land covers in scenario 2, and their corresponding impact on peak runoff rate in site COM.

Land covers	Scenario 1		Scenario 2				
	Coverage (%)	Peak runoff rate (m ³ /s)	Coverage (%)				Peak runoff rate (m ³ /s)
			Trees	Bioswales	Permeable pavements	Green roofs	
Public green spaces	20.18	0.05	0	0	0	0	0.05
Flat roofs	12.02	0.23	0	0	0	12.02	0.08
Sloped roofs	19.87	0.49	0	0	0	0	0.49
Sidewalks	9.32	0.2	4.64	0	4.68	0	0.08
Streets	12.31	0.28	2.32	0.21	9.78	0	0.09
Parking lots	9.07	0.21	1.44	0.14	7.49	0	0.06
Others	17.22	0.4	1.73	0	15.26	0	0.1
Imperviousness	72.60	-	-	-	-	-	-
Perviousness	27.40	-	10.13	0.35	37.21	12.02	-
Total	100	1.86	59.71				0.94
Q Reduction (%)			49.63				

Coverage of Land Use/Land Covers and Peak Runoff Rate Reduction in Site MIX

Site MIX, a typical mixed-land-use area characterized by a combination of residential and commercial properties, features a variety of land covers. These land covers include grasses, tree canopies, and flat and sloped roofs. Sloped roofs accounted for the largest proportion, covering approximately one-third of the total area, while public green spaces accounted for only one-fifth. Flat roofs, sidewalks, and streets covered 7.13 %, 10.47 %, and 13.90 % of the total area. Parking lots represented the smallest percentage of land cover. Overall, the impermeable surfaces

covered three-fourths of the total area and produced a total runoff of 1.82 m³/s, as shown in Table 4.

Given the significant coverage of sidewalks, streets, and parking lots, permeable pavements are the most applicable green infrastructure (GI) feature in scenario 2, representing almost one-third of the total area. Trees are the second most applicable feature at 13.43 %, largely due to their common planting along streets, sidewalks, and parking lots. Green roofs accounted for 7.13% of flat roof installations. Bioswales accounted for only 1% of the total area, applied to streets and parking lots.

Table 4 Distribution of land covers in scenario 1, implementation of GI features across different urban land covers in scenario 2, and their corresponding impact on peak runoff rate in site MIX.

Land covers	Scenario 1		Scenario 2				Peak runoff rate (m³/s)
	Coverage (%)	Peak runoff rate (m³/s)	Coverage (%)				
			Trees	Bioswales	Permeable pavements	Green roofs	
Public green spaces	17.75	0.03	0	0	0	0	0.03
Flat roofs	7.13	0.13	0	0	0	7.13	0.04
Sloped roofs	32.02	0.74	0	0	0	0	0.74
Sidewalks	10.47	0.21	7.78	0	2.69	0	0.1
Streets	13.9	0.3	2.47	0.85	10.58	0	0.09
Parking lots	5.17	0.12	0.09	0.14	4.95	0	0.02
Others	13.55	0.29	3.09	0	10.47	0	0.09
Imperviousness	25.30	-	-	-	-	-	-
Perviousness	74.70	-	13.43	0.99	28.69	7.13	-
Total	100	1.82	50.24				1.11
Q Reduction (%)	38.68						

Implementing 50.24% of GI features reduced runoff to 0.71 m³/s, equivalent to 38.68% for a 2-year storm return period, when impervious cover decreased from three-fourths to one-third.

Discussion

Peak Runoff Rate Reduction in the Two Sites

The two sites were characterized and classified by six primary land covers: public green spaces (parks and gardens), buildings (flat and sloped roofs), sidewalks, streets (roads and boulevards), parking lots, and other categories, such as campuses, driveways, paths, and bare lands. Each site exhibited distinct land cover characteristics: Site COM featured a larger proportion of

public green spaces, flat roofs, and parking lots. Site MIX showed no dominant land cover, with all categories relatively balanced compared to site COM.

The comprehensive peak runoff rate reduction across the two sites resulted from replacing existing impervious surfaces with integrated GI features combined with existing public green spaces. It revealed varying stormwater runoff rates that directly correlated with their distinct land cover compositions. In Figure 3, the significant runoff mitigation at the two sites can be attributed to their land covers, which feature a larger share of other land uses (one-third), followed by streets, parking lots, sidewalks, and flat roofs. These categories present considerable opportunities for the strategic implementation of permeable pavements.

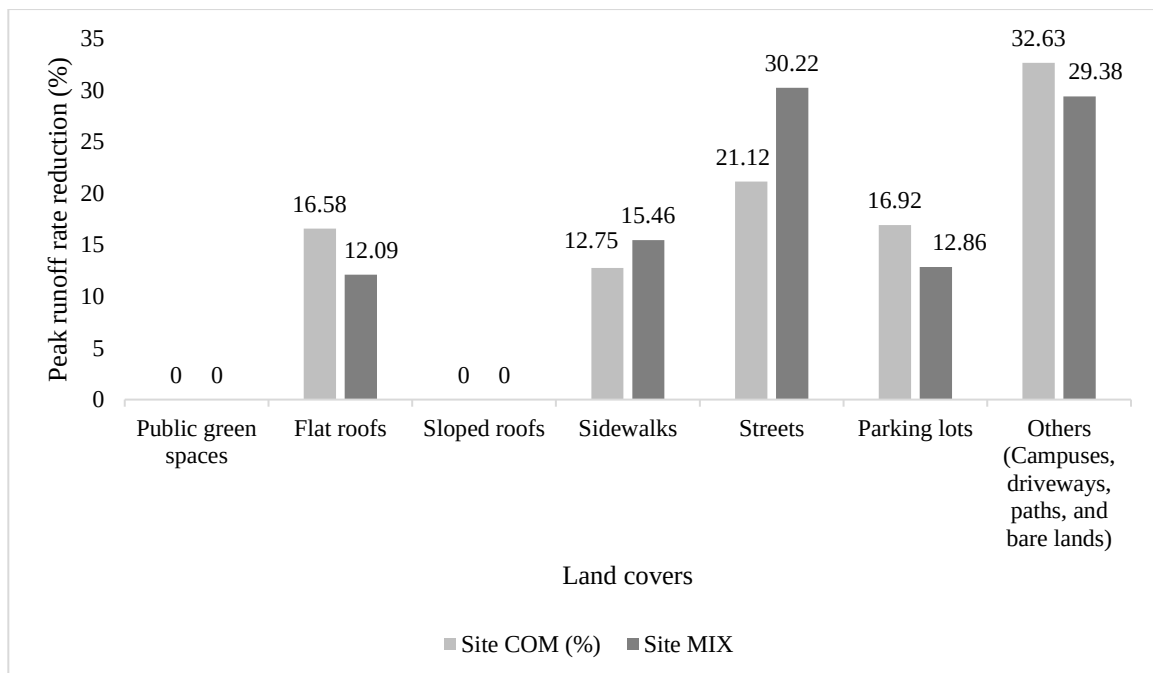


Figure 3 Peak runoff rate reduction of the two sites due to different land covers.

The Effectiveness of Individual GI Features

The effectiveness of individual green infrastructure (GI) features was determined by comparing each feature's coverage to its corresponding peak runoff reduction. This analysis yielded weight values indicating their effectiveness across the three sites. Thanks to Nou and Charoenkit (2020) for providing all the required data on residential land use, referred to as RES in this study.

Table 5 shows the effectiveness of the GI features in minimizing runoff at the sites RES, COM, and MIX. The effectiveness of the GI features in RES was in order: Trees > Permeable pavements = Green roofs > Bioswales. Trees performed the best among the four GI features. It possesses the highest Weight (1.46) since it reduced runoff more than its application. Permeable pavements and Green roofs demonstrated high performance ($0.75 \leq 0.88 < 1$) because they minimized runoff less than their coverage. Conversely, bioswales exhibit the lowest effectiveness (weight < 0.50) due to their limited applicability, resulting in the smallest reduction in runoff.

In site COM, the order was: Permeable pavements > Trees > Green roofs > Bioswales. Permeable pavements and trees were the two best features, as their weights were greater than 1. The underlying cause is their wide application on sidewalks, streets, and other land covers, which collectively account for 80% of the total area. Green roofs have high performance (0.67). While green roofs account for a large share, one-fourth of the combined GI, they only reduced runoff by one-tenth. Bioswales exhibited the lowest effectiveness because their coverage and capacity for runoff reduction were the lowest.

The effectiveness of the GI features was in order: Permeable pavements > Trees > Green roofs > Bioswales. The best feature that weighs more than one is permeable pavements, due to their extensive coverage and high efficiency in reducing runoff. Other features that had high performance were Trees and Green roofs. They approximately reduce runoff by 2:10 (trees) and by 1:10 (green roofs). Bioswales had a moderate performance (0.66) in runoff mitigation due to their limited area of application.

Table 5. Coverage, runoff rate reduction, and Weight of individual GI features in (a), (b), and (c) for site RES, COM, and MIX, respectively.

	Trees			Bioswales			Permeable Pavements			Green Roofs		
	RES	COM	MIX	RES	COM	MIX	RES	COM	MIX	RES	COM	MIX
Coverages (%)	21.64	14.19	21.78	1.08	0.72	2.1	65.3	60.58	60.96	11.98	24.51	15.16
Q_{Reduction} (%)	31.67	15.14	18.98	0.39	0.27	1.39	57.77	68.09	68.31	10.57	16.5	11.32
Weight	1.46	1.07	0.87	0.35	0.37	0.66	0.88	1.12	1.12	0.88	0.67	0.75

Overall, the assessment of individual GI feature effectiveness across the three sites revealed that trees and permeable pavements were the most effective. Trees performed best in sites RES and COM and showed high performance in site MIX. Conversely, Permeable Pavements were most effective at reducing runoff in sites COM and MIX, while also demonstrating high performance in site RES. Green Roofs performed well across the three sites. Bioswales generally performed poorly at sites RES and COM, though they achieved a high level of effectiveness in site MIX for reducing runoff.

The Effectiveness of Combined GI Features

The effectiveness of combined GI features was evaluated by calculating their cumulative Weight and comparing the coverage of their combined GI feature areas to their combined peak runoff rate reduction. The application of 59.71% and 50.24% of combined GI features in site COM and MIX, respectively, to replace impervious covers and to be integrated with existing green spaces, resulted in total peak runoff reductions of 48.86% (0.93 m³/s) and 38.82% (0.70 m³/s), respectively. In RES, Nou and Charoenkit (2020) found that runoff was reduced by 37.90% when 48.14% was provided by GI features. It extracted the largest Weight for combined GI features in site MIX (0.90), followed by site COM (0.86) and site RES (0.65). Consequently, sites COM and MIX demonstrated high performance in reducing stormwater runoff ($0.75 \leq \text{Weight} < 1$), whereas site RES exhibited moderate performance

($0.50 \leq \text{Weight} < 0.75$). This heightened effectiveness in MIX can be attributed to the absence of distinctive dominant land covers, which facilitates the implementation of various individual GI features across approximately equivalent spaces. Although the percentage of permeable pavements, green roofs, and bioswales was higher in MIX than in the other two land uses, their runoff coefficients were lower than those of trees. Consequently, the more extensive coverage of these three features in MIX led to a greater overall runoff reduction compared to COM and RES sites.

This result is comparable with the study by Mei et al. (2018), which conducted an integrated assessment of various GI strategies for flood mitigation within a highly urbanized watershed in Beijing, China, encompassing residential, administrative, and commercial zones. Their findings indicated an impressive peak flow rate reduction of up to 80.62% under a two-year rainfall event, achieved with only 36.59% GI coverage. This substantial reduction in Beijing can be attributed to its temperate monsoon climate, which receives approximately one-third less average annual rainfall than Phnom Penh (522.4 mm). It is important to note that the primary climatic factors influencing runoff volume in tropical regions are high-intensity rainfall, greater runoff-generating capacity, and larger peak flows (Rivard et al., 2006). Despite higher rainfall intensity in tropical climates, evaporation plays a crucial role in hydrologic cycles by reducing runoff, as evaporative potential remains very high throughout the

rainy seasons in tropical monsoon climates (Tsujimoto et al., 2008). This benefit of consistently high temperatures, present for 6 to 9 months per year in tropical regions (Rivard et al., 2006), underscores the substantial impact of GI application on runoff reduction in these climates. The effectiveness of GI is also influenced by the diversity of vegetation species and the characteristics of the substrate or underground soil. For instance, studies on peak runoff reduction in Hong Kong suggest that green roofs with thicker soil layers are more effective in locations experiencing relatively high and prolonged precipitation (annual average precipitation 2,350mm and annual average evapotranspiration 1,123 mm) and high evapotranspiration, with both peak and average runoff reduction increasing proportionally with green roof soil thickness (Liu & Chui, 2019).

Conclusion

This study demonstrates the substantial effectiveness of green infrastructure (GI) in reducing stormwater runoff rates in an urban area of a tropical city. Two typical land-use types in Phnom Penh: commercial (COM) and mixed residential-commercial (MIX) land uses, were selected to investigate in two scenarios. Scenario 1 represented the actual land use/land cover, while Scenario 2 incorporated GI features, including trees, permeable pavements, bioswales, and green roofs. Quantum Geographic Information System (QGIS), supported by Google Satellite overlay imagery, was used to classify land use/land cover and GI features in scenarios 1 and 2, respectively. The Rational Method was employed to estimate

peak runoff rates for both scenarios, and the overall results were computed and analyzed. In Scenario 1, six land cover types were identified: public green spaces (parks and gardens), roofs (flat and sloped), sidewalks, streets (roads and boulevards), parking lots, and other land covers (campuses, driveways, paths, and bare lands).

The inherent characteristics of each land-use type influenced the performance of individual GI features. In the two sites, permeable pavements emerged as the most applicable GI features as they could be applied to four land covers: sidewalks, streets, parking lots, and other land covers. Conversely, bioswales demonstrated slight applicability, as their implementation was limited to boulevards and parking lots, which collectively represent a small share of impervious surfaces. Trees and green roofs were the second and third-most applicable features of the combined GI features, respectively. Overall, the combined GI features covered more than half of the total area at both sites, resulting in runoff reductions of 48.86% and 38.82% for sites COM and MIX, respectively.

Comparing the runoff rate-reduction capacity of GI features and their application yielded insights into their effectiveness. Accordingly, the effectiveness of both individual and combined GI features was obtained for sites COM and MIX. Based on their weighted calculation, trees and permeable pavements exhibited the best results, while green roofs had high performance, and bioswales demonstrated low performance.

In conclusion, this study demonstrates the high performance of green infrastructure (GI)

features integration, such as trees, permeable pavements, bioswales, and green roofs, as a retrofit approach for mitigating urban floods across different land-use/land-cover in central Phnom Penh. To scale up optimal GI performance for runoff reduction in tropical climates, other features such as retention ponds, constructed wetlands, detention ponds, rain gardens, and rainwater harvesting can reduce runoff volume during intense monsoon rains. These features can be combined with distributed and centralized retention strategies to manage both frequent nuisance floods and extreme events, while providing multiple environmental, social, and economic benefits. Future research should quantify the optimal mix of GI and retention for different land-use types and develop design guidelines for integrated green-gray infrastructure in Phnom Penh.

Limitation

It is important to acknowledge that while this study aims to investigate the potential performance of green infrastructure (GI) features in reducing runoff in urban land use, it is subject to a primary limitation. The runoff coefficient values (C and C') and rainfall intensity (i) are derived from the literature, introducing uncertainty in the absolute accuracy of the results. However, the primary focus of this study remains on comparing the relative potential of various GI features for runoff reduction rather than on providing highly precise quantitative estimates. To enhance accuracy in future hydrological studies, the runoff coefficient (C) for given drainage areas should be examined. These values are primarily

contingent on three critical factors: soil properties, land use types, and the catchment's slope, which influence flow characteristics. Furthermore, the runoff coefficient of green infrastructure features (C'), their specific layer criteria, soil composition, vegetation, and the geographical attributes of their respective catchments should be further considered.

Acknowledgement

This study would not be possible without my supervisor's intellectual guidance. I would also like to thank the research and innovation team at CamTech University and the anonymous reviewers.

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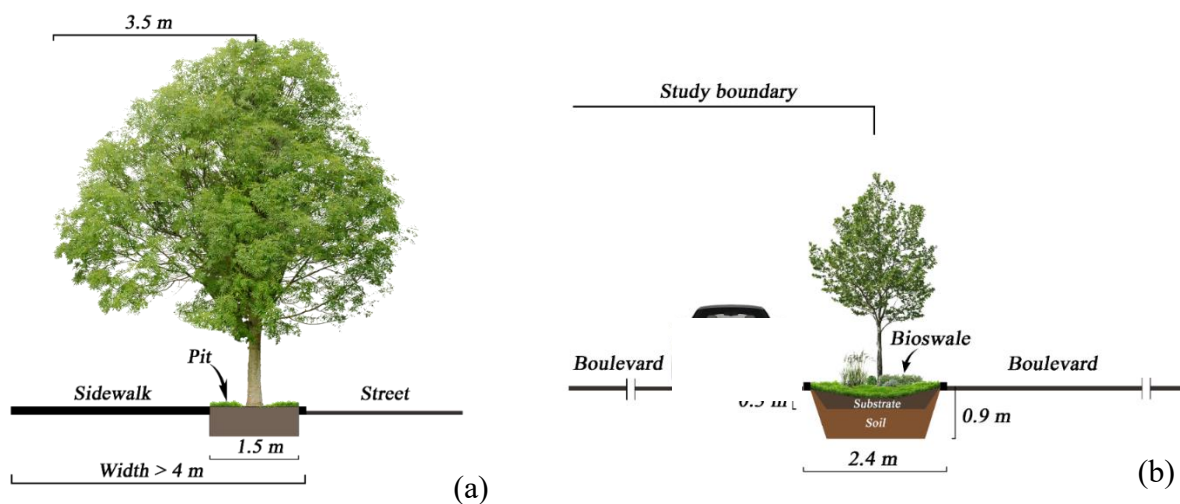
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Appendix A: The specification of a tree on sidewalks, where the width is more than 4 m (type c) in (a), and a bioswale in the median of the boulevard (b). In (a), the average radius of street tree canopies ranged from 1.8 to 5.2 m (at a stem diameter of 25 cm) (Pretzsch et al., 2015). Hence, 1.8 m, 2.5 m, and 3.5 m of tree canopies' radius were employed. And they cover not only sidewalks but also streets.



Appendix B: Summary of GI features' formula of coverage due to their specification for implementation.

GI features	Formula of coverage (m ²)	Specification of the implementation in the various land covers
Trees		- Deciduous tree species - On sidewalks where the widths are wider than 1.5 m, 8m each
Pits	Width x Length	Sidewalks
	0.6 x 0.6	< 2 m
	1 x 1	2 - 4 m
	1.5 x 1.5	> 4 m
Canopies	ΠR^2	It was assumed that tree canopies are two-thirds for sidewalks and over streets (roads and boulevards), with the remaining one-third for the total.
	$(\Pi \times 1.8^2) \times 2/3$	< 2 m Sidewalks
	$(\Pi \times 1.8^2) \times 1/3$	Streets
	$(\Pi \times 2.5^2) \times 2/3$	2- 4 m Sidewalks, radius = 2.5 m
	$(\Pi \times 2.5^2) \times 1/3$	Streets
	$(\Pi \times 3.5^2) \times 2/3$	> 4 m Sidewalks, radius = 3.5 m
	$(\Pi \times 3.5^2) \times 1/3$	Streets
Bioswales	Width x Length	Along the streets (boulevards and roads)
	1.2 x Length	Along boulevards: replaced an existing concrete median barrier - Slope: less than 5% and side slope of 3:1 - Vegetation: grasses, shrubs, and small plants; substrate: 0.3 m; soil depth: 0.9 m (engineered soil) - Width: 2.4 m - Length: more than 2.4 m, otherwise it becomes a tree pit.
	0.6 x Length	In parking lots - Slope: less than 2 % and side slope of 4:1 - Vegetation: grasses and shrubs; substrate: 0.15 m; soil depth: 0.5 m - Width: 0.6 m - Length: more than 0.6 m, otherwise it becomes a tree pit.
Permeable pavements		Sidewalks, parking lots, streets, and others

		The impervious pavers, concrete, and asphalt are replaced by pervious pavers, concrete, and asphalt, respectively.
	$\text{Area}_{\text{sidewalks}} - \text{Area}_{\text{pits}}$	Sidewalks
	$\text{Area}_{\text{parking lots}} - \text{Area}_{\text{bioswales}}$	Parking areas
	$\text{Area}_{\text{streets}} - \text{Area}_{\text{tree canopies}}$	Roads
	$\text{Area}_{\text{streets}} - (\text{Area}_{\text{bioswales}} + \text{Area}_{\text{tree canopies}})$	Boulevards
	$\text{Area}_{\text{others}} - \text{Area}_{\text{existing tree canopies}}$	Others: campuses, driveways, paths, and bare lands.
Green roofs	Area of flat roofs	- Fully covered flat roofs - Intensive green roofs - Key characteristics include vegetation such as sedums, herbs, and small plants, and a substantial substrate depth of 0.15 to 1.6 meters.